Exercise 1D — Full Solutions

Questions 1–8 (complete)

Transforming the roots of an equation by substitution

Colour key: green = verification check | blue = alternative method.

Substitution method. To obtain the equation whose roots are $y = f(x)$, where x runs over the roots of a given equation: rearrange $y = f(x)$ to express x in terms of y, substitute into the given equation, and clear denominators. The resulting polynomial in y has exactly the transformed roots. For roots that are a power of the original ($\alpha^2, \beta^2, \dots$ or $\alpha^3, \beta^3, \dots$) put $x = y^{1/k}$, where $y = x^k$: substituting leaves some terms as polynomials in y while the rest carry a fractional power of y; collect those fractional-index terms on one side and raise to the power k to remove the surd. For power sums of a transformed equation we use the standard recurrence (multiply a root's equation by r^n and sum), with leading term aS_{n+k}.

Question 1

$x^2 + 5x + 3 = 0$ has roots α, β. Find the quadratic with roots $3\alpha, 3\beta$.

Put $y = 3x$, so $x = \dfrac{y}{3}$. Substituting into $x^2 + 5x + 3 = 0$:

$$\frac{y^2}{9} + \frac{5y}{3} + 3 = 0 \xrightarrow{\times 9} \boxed{y^2 + 15y + 27 = 0}.$$

Check. Original: $\alpha + \beta = -5$, $\alpha\beta = 3$. New roots: sum $3(\alpha + \beta) = -15$, product $9\alpha\beta = 27$, giving $y^2 + 15y + 27 = 0$ ✓.

Question 2

$2x^2 - 4x + 7 = 0$ has roots α, β. Find the quadratic with roots **(a)** α^2, β^2; **(b)** $2\alpha - 3, 2\beta - 3$.

(a) Put $y = x^2$, so $x = \sqrt{y} = y^{1/2}$. Substituting into $2x^2 - 4x + 7 = 0$ (with $x^2 = y$) leaves a single surd term:

$$2y - 4\sqrt{y} + 7 = 0.$$

Isolate the fractional-index term on one side and square to remove the surd:

$$4\sqrt{y} = 2y + 7 \implies 16y = (2y + 7)^2 = 4y^2 + 28y + 49 \implies \boxed{4y^2 + 12y + 49 = 0}.$$

Alternative (even/odd split). Write $2x^2 + 7 = 4x$ and square directly to remove x: $(2x^2 + 7)^2 = 16x^2$, i.e. $(2y + 7)^2 = 16y$, giving $4y^2 + 12y + 49 = 0$ as before.

(b) Put $y = 2x - 3$, so $x = \dfrac{y + 3}{2}$. Substituting into $2x^2 - 4x + 7 = 0$:

$$2\left(\frac{y+3}{2}\right)^2 - 4\left(\frac{y+3}{2}\right) + 7 = 0 \implies \frac{(y+3)^2}{2} - 2(y + 3) + 7 = 0.$$

Multiplying by 2 and expanding: $(y + 3)^2 - 4(y + 3) + 14 = 0$, i.e.

$$y^2 + 6y + 9 - 4y - 12 + 14 = 0 \implies \boxed{y^2 + 2y + 11 = 0}.$$

Check. Original: $\alpha+\beta=2,\ \alpha\beta=\frac{7}{2}$. (a) $\sum\alpha^2=(\alpha+\beta)^2-2\alpha\beta=4-7=-3$ and $\alpha^2\beta^2=\frac{49}{4}$, giving $4y^2+12y+49=0$ ✓. (b) sum $2(\alpha+\beta)-6=-2$, product $4\alpha\beta-6(\alpha+\beta)+9=14-12+9=11$ ✓.

Question 3

$3x^2-2x+9=0$ has roots α,β. Find the quadratic with roots $\dfrac{\alpha+1}{\alpha},\dfrac{\beta+1}{\beta}$.

Put $y=\dfrac{x+1}{x}=1+\dfrac{1}{x}$, so $x=\dfrac{1}{y-1}$. Substituting into $3x^2-2x+9=0$:

$$\frac{3}{(y-1)^2}-\frac{2}{y-1}+9=0 \xrightarrow{\times(y-1)^2} 3-2(y-1)+9(y-1)^2=0.$$

Expanding: $3-2y+2+9y^2-18y+9=0$, i.e.

$$\boxed{9y^2-20y+14=0}.$$

Check. Original: $\alpha+\beta=\frac{2}{3},\ \alpha\beta=3$. New roots $1+\frac{1}{\alpha}$: sum $2+\dfrac{\alpha+\beta}{\alpha\beta}=2+\frac{2}{9}=\frac{20}{9}$, product $\left(1+\frac{1}{\alpha}\right)\left(1+\frac{1}{\beta}\right)=1+\frac{2}{9}+\frac{1}{3}=\frac{14}{9}$, giving $9y^2-20y+14=0$ ✓.

Question 4

$x^2-4x+9=0$ has roots α,β. Find the quadratic with roots $\dfrac{1}{\alpha},\dfrac{1}{\beta}$.

Put $y=\dfrac{1}{x}$, so $x=\dfrac{1}{y}$. Substituting into $x^2-4x+9=0$:

$$\frac{1}{y^2}-\frac{4}{y}+9=0 \xrightarrow{\times y^2} 1-4y+9y^2=0 \implies \boxed{9y^2-4y+1=0}.$$

Check. Original: $\alpha+\beta=4,\ \alpha\beta=9$. New roots: sum $\dfrac{1}{\alpha}+\dfrac{1}{\beta}=\dfrac{\alpha+\beta}{\alpha\beta}=\dfrac{4}{9}$ and product $\dfrac{1}{\alpha\beta}=\dfrac{1}{9}$, giving $y^2-\frac{4}{9}y+\frac{1}{9}=0$, i.e. $9y^2-4y+1=0$ ✓.

Question 5

$2x^3-5x+1=0$ has roots α,β,γ. Find the cubic with roots $\alpha^2,\beta^2,\gamma^2$, and hence find S_4.

Put $y=x^2$, so $x=\sqrt{y}=y^{1/2}$. Substituting into $2x^3-5x+1=0$, with $x^3=x\cdot x^2=y\sqrt{y}$, the even part is constant and the odd part carries $\sqrt{y}$:

$$2y\sqrt{y}-5\sqrt{y}+1=0.$$

Collect the fractional-index terms on one side and square:

$$\sqrt{y}\,(2y-5)=-1 \implies y(2y-5)^2=1 \implies y(4y^2-20y+25)=1 \implies \boxed{4y^3-20y^2+25y-1=0}.$$

Alternative (isolate the odd part). From $2x^3 - 5x + 1 = 0$, $x(2x^2 - 5) = -1$, i.e. $x(2y - 5) = -1$, so $x = \dfrac{-1}{2y-5}$. Then $x^2 = y$ gives $\dfrac{1}{(2y-5)^2} = y$, i.e. $y(2y-5)^2 = 1$, the same cubic.

Hence S_4: the new roots are $\alpha^2, \beta^2, \gamma^2$, so $\sum \alpha^4 = S_4$ is the *sum of squares* of the new roots. From $4y^3 - 20y^2 + 25y - 1 = 0$, $\sum \alpha^2 = \dfrac{20}{4} = 5$ and $\sum \alpha^2\beta^2 = \dfrac{25}{4}$, so

$$S_4 = \left(\sum \alpha^2\right)^2 - 2\sum \alpha^2\beta^2 = 5^2 - 2\left(\tfrac{25}{4}\right) = 25 - \tfrac{25}{2} = \tfrac{25}{2}.$$

Cross-check via the recurrence on the original. For $2x^3 - 5x + 1 = 0$ ($a = 2, b = 0, c = -5, d = 1$): $2S_{n+3} - 5S_{n+1} + S_n = 0$, $S_0 = 3$, with $S_1 = 0$ and $S_2 = 0 - 2(-\frac{5}{2}) = 5$. Then $n = 0$: $2S_3 + S_0 = 0 \Rightarrow S_3 = -\frac{3}{2}$; $n = 1$: $2S_4 - 5S_2 = 0 \Rightarrow S_4 = \frac{25}{2}$.

Check. Numerically the new cubic has root-sum 5.000, and $S_4 = 12.500 = \frac{25}{2}$ ✓.

Question 6

$x^3 + 3x^2 - 1 = 0$ has roots α, β, γ. Show that the cubic with roots $\dfrac{\alpha+2}{\alpha}, \dfrac{\beta+2}{\beta}, \dfrac{\gamma+2}{\gamma}$ is $y^3 - 3y^2 - 9y + 3 = 0$; hence find **(a)** $\dfrac{(\alpha+2)(\beta+2)(\gamma+2)}{\alpha\beta\gamma}$ and **(b)** $\dfrac{\alpha}{\alpha+2} + \dfrac{\beta}{\beta+2} + \dfrac{\gamma}{\gamma+2}$.

Put $y = \dfrac{x+2}{x} = 1 + \dfrac{2}{x}$, so $x = \dfrac{2}{y-1}$. Substituting into $x^3 + 3x^2 - 1 = 0$:

$$\frac{8}{(y-1)^3} + \frac{12}{(y-1)^2} - 1 = 0 \xrightarrow{\times (y-1)^3} 8 + 12(y-1) - (y-1)^3 = 0.$$

Since $(y-1)^3 = y^3 - 3y^2 + 3y - 1$,

$$8 + 12y - 12 - \left(y^3 - 3y^2 + 3y - 1\right) = 0 \implies -y^3 + 3y^2 + 9y - 3 = 0 \implies y^3 - 3y^2 - 9y + 3 = 0,$$

as required. For this cubic: $\sum y = 3$, $\sum y_iy_j = -9$, $y_1y_2y_3 = -3$.

(a) $\dfrac{(\alpha+2)(\beta+2)(\gamma+2)}{\alpha\beta\gamma} = \dfrac{\alpha+2}{\alpha} \cdot \dfrac{\beta+2}{\beta} \cdot \dfrac{\gamma+2}{\gamma}$ is the product of the new roots $= y_1y_2y_3 = -3$.

(b) $\dfrac{\alpha}{\alpha+2} = \dfrac{1}{(\alpha+2)/\alpha} = \dfrac{1}{y}$, so the sum is

$$\sum \frac{1}{y} = \frac{\sum y_iy_j}{y_1y_2y_3} = \frac{-9}{-3} = 3.$$

Check. Numerically the product in (a) evaluates to -3.000 and the sum in (b) to 3.000 ✓.

Question 7

$2x^4 - x^3 - 6 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Show that the quartic with roots $\alpha^3, \beta^3, \gamma^3, \delta^3$ is $8y^4 - y^3 - 18y^2 - 108y - 216 = 0$; hence find S_6 and S_{-3}.

Put $y = x^3$, so $x = y^{1/3}$. Substituting into $2x^4 - x^3 - 6 = 0$, with $x^3 = y$ and $x^4 = x \cdot x^3 = y \cdot y^{1/3} = y^{4/3}$, the surd appears only in the leading term:

$$2y^{4/3} - y - 6 = 0.$$

Isolate the fractional-index term and cube to remove the cube root (note $(y^{4/3})^3 = y^4$):

$2y^{4/3} = y+6 \implies 8y^4 = (y+6)^3 = y^3+18y^2+108y+216 \implies 8y^4-y^3-18y^2-108y-216=0,$

as required. For this quartic ($a=8, b=-1, c=-18, d=-108, e=-216$):

Alternative (substitute $x = \frac{y+6}{2y}$). Write $x^4 = x\cdot x^3 = xy$, giving $2xy - y - 6 = 0$, so $x = \dfrac{y+6}{2y}$. Then $x^3 = y$ gives $\dfrac{(y+6)^3}{8y^3} = y$, i.e. $(y+6)^3 = 8y^4$, the same quartic.

$$\sum\alpha^3 = -\frac{b}{a} = \frac{1}{8}, \quad \sum\alpha^3\beta^3 = \frac{c}{a} = -\frac{9}{4}, \quad \sum\alpha^3\beta^3\gamma^3 = -\frac{d}{a} = \frac{27}{2}, \quad \alpha^3\beta^3\gamma^3\delta^3 = \frac{e}{a} = -27.$$

S_6**:** $S_6 = \sum\alpha^6$ is the sum of squares of the new roots:

$$S_6 = \left(\sum\alpha^3\right)^2 - 2\sum\alpha^3\beta^3 = \left(\tfrac{1}{8}\right)^2 - 2\left(-\tfrac{9}{4}\right) = \tfrac{1}{64} + \tfrac{9}{2} = \tfrac{289}{64}.$$

S_{-3}**:** $S_{-3} = \sum\dfrac{1}{\alpha^3} = \dfrac{\sum\alpha^3\beta^3\gamma^3}{\alpha^3\beta^3\gamma^3\delta^3} = \dfrac{27/2}{-27} = -\dfrac{1}{2}.$

Check. Numerically $S_6 = 4.51563 = \frac{289}{64}$ and $S_{-3} = -0.500$ ✓.

Question 8

$x^3-x+4=0$ has roots α,β,γ. Find the cubic with roots $\alpha^2,\beta^2,\gamma^2$, and hence determine S_6, S_8, S_{10}.
Put $y = x^2$, so $x = \sqrt{y} = y^{1/2}$. Substituting into $x^3-x+4=0$, with $x^3 = y\sqrt{y}$, the surd appears only in the odd part:

$$y\sqrt{y} - \sqrt{y} + 4 = 0.$$

Collect the fractional-index terms on one side and square:

$$\sqrt{y}\,(y-1) = -4 \implies y(y-1)^2 = 16 \implies \boxed{y^3-2y^2+y-16=0}.$$

Alternative (factorise then substitute). From $x^3-x+4=0$, $x(x^2-1) = -4$, i.e. $x(y-1) = -4$, so $x = \dfrac{-4}{y-1}$. Then $x^2 = y$ gives $\dfrac{16}{(y-1)^2} = y$, i.e. $y(y-1)^2 = 16$, the same cubic.

Hence the power sums. The new roots are $\alpha^2,\beta^2,\gamma^2$, so their power sums $P_k = \sum(\alpha^2)^k = \sum\alpha^{2k} = S_{2k}$. Thus $S_6 = P_3,\ S_8 = P_4,\ S_{10} = P_5$ for the cubic $y^3-2y^2+y-16=0$ ($a=1, b=-2, c=1, d=-16$).
Recurrence (multiply a root's equation by y^n and sum): $P_{n+3} - 2P_{n+2} + P_{n+1} - 16P_n = 0,\ P_0 = 3.$
Seeds:

$$P_1 = \sum\alpha^2 = 2, \qquad P_2 = \left(\sum\alpha^2\right)^2 - 2\sum\alpha^2\beta^2 = 2^2 - 2(1) = 2.$$

$$n=0:\ P_3 = 2P_2 - P_1 + 16P_0 = 4 - 2 + 48 = 50 = S_6,$$

$$n=1:\ P_4 = 2P_3 - P_2 + 16P_1 = 100 - 2 + 32 = 130 = S_8,$$

$$n=2:\ P_5 = 2P_4 - P_3 + 16P_2 = 260 - 50 + 32 = 242 = S_{10}.$$

Check. Numerically $S_6 = 50.000,\ S_8 = 130.000,\ S_{10} = 242.000$ ✓.